

ELECTRIC SUBMERSIBLE PUMP (ESP) DESIGN OF A VSD INSTALLATION

TUGAS AKHIR SEMESTER



Oleh :

Nama Mahasiswa	: Larasati Kris Sadewi Rahma Musyafia Muhammad
NIM	: 15412012 15412024
Program Studi	: Teknik Produksi Migas
Konsentrasi	: Produksi
Diploma	: III (Tiga)

**KEMENTERIAN ENERGI DAN SUMBER DAYA MINERAL
BADAN PENGEMBANGAN SUMBER DAYA MANUSIA ENERGI DAN SUMBER DAYA MINERAL
SEKOLAH TINGGI ENERGI DAN MINERAL Akamigas**

Cepu, Desember 2017

I. Data

Well Physical Data

Casing ID	6.331 in
Tubing ID	2.441 in
Tubing OD	2.875 in
Tubing New	YES
Perforation Depth	6,500 ft
Pump Depth	6,000 ft

Well Performance Data

WHP @desired rates	100 psi
CHP @desired rates	20 psi
Min. desired liquid rate @a water cut of	2,000 STB/d 50%
Max. desired liquid rate @a water cut of	2,500 STB/d 60%
SBHP	2,200 psi
PI	1.5 bpd/psi
GOR produced	300 scf/STB
BHT	150 °F

Fluid Properties

Sp. Gr. of oil	0.86
Sp. Gr. of water	1.08
Sp. Gr. of gas	0.7

**Power supply data:* Available surface voltage 4,500 volts

II. Solution

A. Well Inflow Calculation for The Minimum Rate

1. Flowing Bottom Hole Pressure (FBHP) at the minimum rate

$$FBHP = SBHP - \frac{\text{Min. desired liquid rate}}{PI}$$

$$FBHP = 2,200 - \frac{2000}{1.5}$$

$$FBHP = 867 \text{ psi}$$

2. Pump Intake Pressure (PIP) Calculation

$$PIP = P_{wf} - i$$

$$PIP = 660 \text{ psi}$$

3. Determine API gravity

$$^{\circ}API = \frac{141.5}{SG_o} - 131.5$$

$$^{\circ}API = \frac{141.5}{0.86} - 131.5$$

$$^{\circ}API = 33$$

4. Determine variable y

$$y = (0.00091 \times T) - (0.01215 \times ^{\circ}API)$$

$$y = (0.00091 \times 150) - (0.01215 \times 33)$$

$$y = -0.264$$

5. Determine Rs. From y calculation it can find Rs value

$$Rs = Sg_{gas} \left[\left(\frac{PIP}{18 \times 10^y} \right) \right]^{1.205}$$

$$Rs = 0.7 \left[\left(\frac{660}{18 \times 10^{-0.264}} \right) \right]^{1.205}$$

$$R_s = 116 \text{ scf/bbl}$$

6. Ppc and Tpc Calculation

$$P_{pc} = 709 - 58.7 \times SG_{gas}$$

$$P_{pc} = 709 - 58.7 \times 0.7$$

$$P_{pc} = 668 \text{ psia}$$

$$T_{pc} = 170.5 - 307.3 \times SG_{gas}$$

$$T_{pc} = 170.5 - 307.3 \times 0.7$$

$$T_{pc} = 386 \text{ R}$$

7. Ppr and Tpr Calculation

$$P_{pr} = \frac{PIP + 14.7}{P_{pc}}$$

$$P_{pr} = \frac{660 + 14.7}{668}$$

$$P_{pr} = 1.01$$

$$T_{pr} = \frac{T + 460}{T_{pc}}$$

$$T_{pr} = \frac{150 + 460}{386}$$

$$T_{pr} = 1.58$$

8. From Ppr and Tpr Calculation, variable Z can be known, here is how to determine the variable Z

$$Z = 1 - \left(A \times \frac{P_{pr}}{10^{0.9813}} \right)^{T_{pr}} + \left(0.274 \times \frac{(P_{pr})^2}{10^{0.8157}} \right)^{1.58}$$

$$Z = 1 - \left(3.52 \times \frac{P_{pr}}{10^{0.9813}} \right)^{T_{pr}} + \left(0.274 \times \frac{(P_{pr})^2}{10^{0.8157}} \right)^{1.58}$$

$$Z=0.915$$

9. Now gas volume factor can be determined from the following calculation

$$Bg = 0.0283 \times Z \times \left[\frac{T + 460}{PIP + 14.7} \right]$$

$$Bg = 0.0283 \times 0.915 \times \left[\frac{150 + 460}{660 + 14.7} \right]$$

$$Bg = 0.023 \text{ ft}^3/\text{scf}$$

10. Volumetric rate of free gas is found from the following calculation

$$Qg' = Qo \times (GOR - Rs) \times Bg$$

$$Qg' = 1000 \times (300 - 116) \times 0.023$$

$$Qg' = 4,232 \text{ ft}^3/\text{d}$$

11. To find the liquid volumetric rate, the determination of the oil volume factor can be found from the following calculation. Determine variable F

$$F' = Rs \times \left(\frac{Sg \text{ gas}}{Sg \text{ oil}} \right)^{0.5} + 1.25 \times T$$

$$F' = 116 \times \left(\frac{0.7}{0.86} \right)^{0.5} + 1.25 \times 150$$

$$F' = 292$$

$$Bo = 0.972 + 0.000147 \times F^{1.175}$$

$$Bo = 0.972 + 0.000147 \times 292^{1.175}$$

$$Bo = 1.088 \text{ bbl/STB}$$

12. The liquid volumetric rate handled by the pump is found by the following calculation and assuming $B_w = 1$

$$$$

$$Ql = (Q_o \times B_o) + (Q_w \times B_w)$$

$$Ql = (1,000 \times 1.088) + (1,000 \times 1)$$

$$Ql = 2,088 \text{ bpd}$$

B. Well Inflow Calculation for The Maximum Rate

1. Flowing Bottom Hole Pressure (FBHP) at the minimum rate

$$FBHP = SBHP - \frac{\text{Min. desired liquid rate}}{PI}$$

$$FBHP = 2,200 - \frac{2500}{1.5}$$

$$FBHP = 533 \text{ psi}$$

2. Pump Intake Pressure (PIP) Calculation

$$PIP = P_{wf} - i$$

$$PIP = 330 \text{ psi}$$

3. Determine API gravity

$$^{\circ}API = \frac{141.5}{SG_o} - 131.5$$

$$^{\circ}API = \frac{141.5}{0.86} - 131.5$$

$$^{\circ}API = 33$$

4. Determine variable y

$$y = (0.00091 \times T) - (0.01215 \times ^{\circ}API)$$

$$y = (0.00091 \times 150) - (0.01215 \times 33)$$

$$y = -0.264$$

5. Determine Rs. From y calculation it can find Rs value

$$R_s = Sg_{gas} \left[\left(\frac{PIP}{18 \times 10^y} \right) \right]^{1.205}$$

$$R_s = 0.7 \left[\left(\frac{330}{18 \times 10^{-0.264}} \right) \right]^{1.205}$$

$$R_s = 50 \text{ scf/bbl}$$

6. Ppc and Tpc Calculation

$$P_{pc} = 709 - 58.7 \times SG_{gas}$$

$$P_{pc} = 709 - 58.7 \times 0.7$$

$$P_{pc} = 668 \text{ psia}$$

$$T_{pc} = 170.5 - 307.3 \times SG_{gas}$$

$$T_{pc} = 170.5 - 307.3 \times 0.7$$

$$T_{pc} = 386 \text{ R}$$

7. Ppr and Tpr Calculation

$$P_{pr} = \frac{PIP + 14.7}{P_{pc}}$$

$$P_{pr} = \frac{330 + 14.7}{668}$$

$$P_{pr} = 0.516$$

$$T_{pr} = \frac{T + 460}{T_{pr}}$$

$$T_{pr} = \frac{150 + 460}{386}$$

$$T_{pr} = 1.58$$

8. From Ppr and Tpr Calculation, variable Z can be known, here is how to determine the variable Z

$$Z = 1 - \left(A \times \frac{P_{pr}}{10^{0.9813}} \right)^{T_{pr}} + \left(0.274 \times \frac{(P_{pr})^2}{10^{0.8157}} \right)^{1.58}$$

$$Z = 1 - \left(3.52 \times \frac{P_{pr}}{10^{0.9813}} \right)^{T_{pr}} + \left(0.274 \times \frac{(P_{pr})^2}{10^{0.8157}} \right)^{1.58}$$

$$Z = 0.953$$

9. Now gas volume factor can be determined from the following calculation

$$Bg = 0.0283 \times Z \times \left[\frac{T + 460}{PIP + 14.7} \right]$$

$$Bg = 0.0283 \times 0.955 \times \left[\frac{150 + 460}{330 + 14.7} \right]$$

$$Bg = 0.048 \text{ ft}^3/\text{scf}$$

10. Volumetric rate of free gas is found from the following calculation

$$Qg' = Qo \times (GOR - Rs) \times Bg$$

$$Qg' = (2500 \times 0.4) \times (300 - 50) \times 0.048$$

$$Qg' = 12,000 \text{ ft}^3/\text{d}$$

11. To find the liquid volumetric rate, the determination of the oil volume factor can be found from the following calculation. Determine variable F

$$$$

$$F' = R_s \times \left(\frac{Sg_{gas}}{Sg_{oil}} \right)^{0.5} + 1.25 \times T$$

$$F' = 50 \times \left(\frac{0.7}{0.86} \right)^{0.5} + 1.25 \times 150$$

$$F' = 232$$

$$Bo = 0.972 + 0.000147 \times F'^{1.175}$$

$$Bo = 0.972 + 0.000147 \times 232^{1.175}$$

$$Bo = 1.061 \text{ bbl/STB}$$

12. The liquid volumetric rate handled by the pump is found by the following calculation and assuming $B_w = 1$

$$Q_l = (Q_o \times Bo) + (Q_w \times B_w)$$

$$Q_l = (2,500 \times 0.4 \times 1.061) + (2,500 \times 0.6)$$

$$Q_l = 2,561 \text{ bpd}$$

C. Natural Gas Separation Calculation for The Minimum Rate

1. Calculation of the cross-sectional area of the annulus

$$A = 0.0055 \times (Casing \text{ ID}^2 - Tubing \text{ OD}^2)$$

$$A = 0.0055 \times (6.331^2 - 2.875^2)$$

$$A = 0.175 \text{ ft}^2$$

2. The liquid superficial velocity in the annulus is calculated with $WOR = 1$ and $B_w = 1$ as:

$$V_{sl} = 6.5 \times 10^{-5} \times \left(\frac{Q_l}{A} \right) \times \left(\frac{Bo}{2} + \frac{1}{2} \right)$$

$$$$

$$V_{sl} = 6.5 E - 5 \times \left(\frac{2000}{0.175} \right) \times \left(\frac{1.088}{2} + \frac{1}{2} \right)$$

$$V_{sl} = 0.78 \text{ ft/sec}$$

3. Calculation of density of gas phase at suction conditions

$$\rho_g = 0.0764 \times \left(\frac{SG_{gas}}{Bg} \right)$$

$$\rho_g = 0.0764 \times \left(\frac{0.7}{0.023} \right)$$

$$\rho_g = 2.3 \text{ lb/ft}^3$$

4. Calculation the density of liquid phase

$$\rho_l = 62.5 \times \left(\frac{0.86}{Bo \times 2} \right) + \left(\frac{1.08}{2} \right)$$

$$\rho_l = 62.5 \times \left(\frac{0.86}{1.088 \times 2} \right) + \left(\frac{1.08}{2} \right)$$

$$\rho_l = 58.5 \text{ lb/ft}^3$$

5. Calculation for terminal rising velocity of the gas bubbles

$$V_b = 2 \frac{1}{2} \left\{ \frac{(0.04 \times 32 \times (\rho_l - \rho_g))}{(\rho_l)^2} \right\}^{1.4}$$

$$V_b = 2 \frac{1}{2} \left\{ \frac{(0.04 \times 32 \times (58.5 - 2.3))}{(58.5)^2} \right\}^{1.4}$$

$$V_b = 0.539 \text{ ft/sec}$$

6. The natural gas separation efficiency of the annulus is calculated using the following calculation:

$$\eta n = \left[\frac{V_b}{V_b + V_{sl}} \right] \times 100$$

$$\eta n = \left[\frac{0.539}{0.539 + 0.78} \right] \times 100$$

$$\eta n = 41 \%$$

7. The amount of gas entering the pump can be found from the following calculation

$$Q_{ing} = \left(\frac{Q_{gas}}{5.61} \right) \times (1 - \eta n)$$

$$Q_{ing} = \left(\frac{4,232}{5.61} \right) \times (1 - 0.41)$$

$$Q_{ing} = 454 \text{ bpd}$$

8. The Turpin parameter Φ can be determined by the following calculation

$$\Phi = \frac{Q_{l \text{ minimum desired}} \times Q_{ing}}{3 \times PIP \times Q_l}$$

$$\Phi = \frac{2000 \times 454}{3 \times 660 \times 2,088}$$

$$\Phi = 0.22 < 1$$

No gas separation is required

D. Natural gas separation calculation for the maximum rate

1. Calculation of the cross-sectional area of the annulus

$$A = 0.0055 \times (Casing\ ID^2 - Tubing\ OD^2)$$

$$A = 0.0055 \times (6.331^2 - 2.875^2)$$

$$A = 0.175\ ft^2$$

2. The liquid superficial velocity in the annulus is calculated with $WOR = 1$ and $B_w = 1$ as:

$$V_{sl} = 6.5 \times 10^{-5} \times \left(\frac{Q_l}{A} \right) \times \left(\frac{Bo}{2} + \frac{1}{2} \right)$$

$$V_{sl} = 6.5 \times 10^{-5} \times \left(\frac{2500}{0.175} \right) \times \left(\frac{1.061}{2} + \frac{1}{2.5} \right)$$

$$V_{sl} = 0.95\ ft/sec$$

3. Calculation of density of gas phase at suction conditions

$$\rho_g = 0.0764 \times \left(\frac{SG_{gas}}{Bg} \right)$$

$$\rho_g = 0.0764 \times \left(\frac{0.7}{0.048} \right)$$

$$\rho_g = 1.1\ lb/ft^3$$

4. Calculation the density of liquid phase

$$\rho_l = 62.5 \times \left(\frac{0.86}{Bo \times 2} \right) + \left(\frac{1.08}{2} \right)$$

$$\rho_l = 62.5 \times \left(\frac{0.86}{1.061 \times 2.5} \right) + \left(\frac{1.08}{2.5} \right)$$

$$\rho_l = 60.8\ lb/ft^3$$

5. Calculation for terminal rising velocity of the gas bubbles

$$Vb = 2 \frac{1}{2} \left\{ \left[\frac{(0.04 \times 32 \times (\rho_1 - \rho_g))}{(\rho_1)^2} \right] \right\}^{1.4}$$

$$Vb = 2 \frac{1}{2} \left\{ \left[\frac{(0.04 \times 32 \times (60.8 - 1.1))}{(60.8)^2} \right] \right\}^{1.4}$$

$$Vb = 0.536 \text{ ft/sec}$$

6. The natural gas separation efficiency of the annulus is calculated using the following calculation:

$$\eta_n = \left[\frac{Vb}{Vb + V_{sl}} \right] \times 100$$

$$\eta_n = \left[\frac{0.536}{0.536 + 0.95} \right] \times 100$$

$$\eta_n = 36\%$$

7. The amount of gas entering the pump can be found from the following calculation

$$Q_{ing} = \left(\frac{Q_{gas}}{5.61} \right) \times (1 - \eta_n)$$

$$Q_{ing} = \left(\frac{12,000}{5.61} \right) \times (1 - 0.36)$$

$$Q_{ing} = 1,368 \text{ bpd}$$

8. The Turpin parameter Φ can be determined by the following calculation

$$\Phi = \frac{Ql_{\text{minimum desired}} \times Q_{ing}}{3 \times PIP \times Ql}$$

$$\Phi = \frac{2000 \times 1,368}{3 \times 330 \times 2,561}$$

$$\Phi = 1,08 > 1$$

Gas separation is required

E. Rotary gas separation for the minimum rate

Let us assume that a properly selected rotary gas separator achieves a separation efficiency of $\eta_{sep} = 60\%$. The total combined volume of gas ingested by the pump is determined by the following calculation:

$$Q_{ing} = \left(\frac{Q_{gas}}{5.61} \right) \times (1 - \eta_n) \times (1 - \eta_{sep})$$

$$Q_{ing} = \left(\frac{4,232}{5.61} \right) \times (1 - 0.41) \times (1 - 0.60)$$

$$Q_{ing} = 182 \text{ bpd}$$

The total in-situ fluid volume the ESP pump must handle is the sum of the liquid and gas rates, so we can calculate the total Q_{liquid}

$$Q_{liq} = Ql + Q_{ing}(\text{new})$$

$$Q_{liq} = 2,088 + 182$$

$$Q_{liq} = 2,270 \text{ bpd}$$

The Turpin parameter Φ can be calculated by the following calculation

$$\Phi = \frac{Ql_{\text{minimum desired}} \times Q_{\text{ing}}(\text{new})}{3 \times PIP \times Ql(\text{new})}$$

$$\Phi = \frac{2000 \times 182}{3 \times 660 \times 2,270}$$

$$\Phi = 0.08 < 1$$

Operation of the pump is stable now.

F. Rotary gas separation for the maximum rate

If a rotary gas separator achieves a separation efficiency of $\eta_{\text{sep}} = 60\%$, then the total combined volume of gas ingested by the pump can be determined by the following calculation

$$Q_{\text{ing}} = \left(\frac{Q_{\text{gas}}}{5.61} \right) \times (1 - \eta_n) \times (1 - \eta_{\text{sep}})$$

$$Q_{\text{ing}} = \left(\frac{12,000}{5.61} \right) \times (1 - 0.36) \times (1 - 0.60)$$

$$Q_{\text{ing}} = 547 \text{ bpd}$$

The total in-situ fluid volume the ESP pump must handle is the sum of the liquid and gas rates, so we can calculate the total Q_{liquid}

$$Q_{\text{liq}} = Ql + Q_{\text{ing}}(\text{new})$$

$$Q_{\text{liq}} = 2,561 + 547$$

$$Q_{\text{liq}} = 3,108 \text{ bpd}$$

The Turpin parameter Φ can be determined by the following calculation

$$\Phi = \frac{Ql_{\text{minimum desired}} \times Q_{\text{ing}}(\text{new})}{3 \times PIP \times Ql(\text{new})}$$

$$\Phi = \frac{2,000 \times 547}{3 \times 330 \times 3,108}$$

$$\Phi = 0.36 < 1$$

Now the operation of the pump is stable.

The Turpin parameter Φ can be calculated by the following calculation

$$\Phi = \frac{Ql_{\text{minimum desired}} \times Q_{\text{ing}}(\text{new})}{3 \times PIP \times Ql(\text{new})}$$

$$\Phi = \frac{2000 \times 182}{3 \times 660 \times 2,270}$$

$$\Phi = 0.08 < 1$$

G. TDH Calculation for the minimum rate

In order to find the pump's required discharge pressure, P_d , vertical multi-phase pressure drop calculations must be performed in the tubing string from the wellhead to the pump setting depth. The gas/liquid ratio to be used can be found by the following calculation

$$GLR = \left\{ \left[\frac{(5.61 \times 182)}{Bg} \right] + \left[\frac{Rs \times Qg}{Ql_{\text{minimum desired}}} \right] \right\}$$

$$GLR = \left\{ \left[\frac{(5.61 \times 182)}{0.023} \right] + \left[\frac{116 \times 1,000}{2,000} \right] \right\}$$

$$GLR = 80 \text{ scf/STB}$$

The required pump discharge can be determined by the following calculation

$$Pd = FBHP - (L_{perfo} - L_{set}) \times (Gf + \Delta P_{pump})$$

$$Pd = 2,400 \text{ psi}$$

The required TDH is now found from the pressure increase required and the liquid's density, so here is the calculation

$$TDH = 144 \times \frac{Pd - PIP}{\rho \cdot 1}$$

$$TDH = 144 \times \frac{2,400 - 660}{58.5}$$

$$TDH = 4,283 \text{ ft}$$

H. TDH Calculation for the maximum rate

In order to find the pump's required discharge pressure, Pd, vertical multi-phase pressure drop calculations must be performed in the tubing string from the wellhead to the pump setting depth. The gas/liquid ratio to be used can be found by the following calculation

$$GLR = \left\{ \left[\frac{(5.61 \times 182)}{Bg} \right] + \left[\frac{Rs \times Qg}{Ql_{\text{minimum desired}}} \right] \right\}$$

$$GLR = \left\{ \left[\frac{(5.61 \times 547)}{0.048} \right] + \left[\frac{50 \times 2,500 \times 0.4}{2,500} \right] \right\}$$

$$GLR = 46 \text{ scf / STB}$$

The required pump discharge can be determined by the following calculation

$$$$

$$Pd = FBHP - (L_{perfo} - L_{set}) \times (Gf + \Delta P_{pump})$$

$$Pd = 2600 \text{ psi}$$

The required TDH is now found from the pressure increase required and the liquid's density, so here is the calculation

$$TDH = 144 \times \frac{Pd - PIP}{\rho \cdot 1}$$

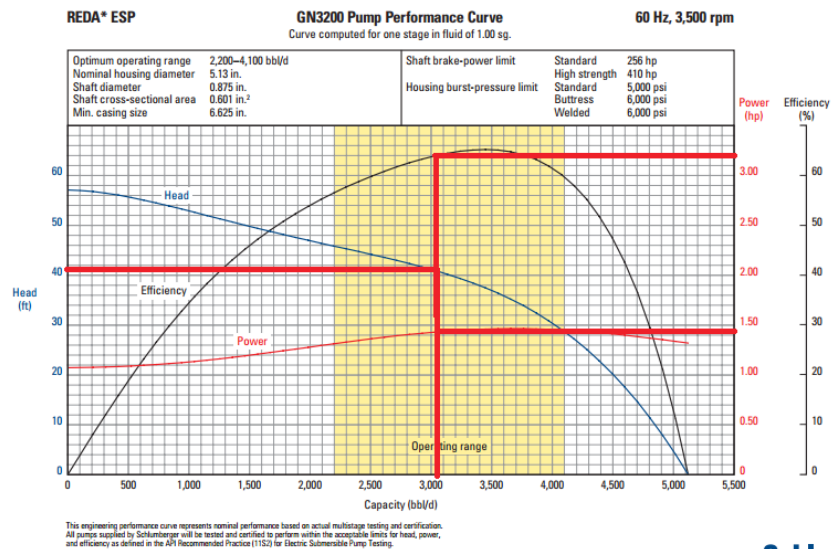
$$TDH = 144 \times \frac{2,600 - 330}{60.8}$$

$$TDH = 5,376 \text{ ft}$$

I. Selection of The Pump for The Maximum Rate

Based on the casing, tubing size and the liquid rate, 540 Series GN33200 is required for this well.

GN3200 Pump



Schlumberger

Required frequency

60Hz

Head/stage	41 Head/stage
HP/stage	1.5 HP/stage
BEP	3,200 bfpd
Range	2,200 bfpd – 4,100 bfpd

Stages

$$\text{Stages} = \frac{\text{Head}}{\text{Head/stages}}$$

$$\text{Stages} = \frac{5,376}{41}$$

$$\text{Stages} = 132 \text{ stages}$$

Determine HP that are needed for one stage of the pump

$$\text{HP} = \text{HP} \times \text{stages}$$

$$\text{HP} = 1.5 \times 132$$

$$\text{HP} = 198 \text{ HP}$$

***From GN3200 Catalog this pump will require 2 EA stages. 1 EA 40 stages – 6.3 ft long and 1 EA 100 stages 14.7 ft long. This pump will be made up in tandem.**

Selecting Protector

With the well conditions, the best protector is BPBSL 540 Series-2550 NTB-1HP.

Selecting VGSA

It requires 540 Series, VGSA S20-90, 6HP

Selecting AGH

It requires 540 Series, AGH G20-40, 37.5HP

Motor Selection

Before choose the motor we need to calculate total HP that will needed for the motor. Total HP is the sum from Pump HP, VGSA HP, AGH HP, and Protector HP

$$\text{Total HP} = \text{Pump HP} + \text{VGSA HP} + \text{AGH HP} + \text{Protector}$$

$$\text{Total HP} = 198 + 6 + 37.5 + 1$$

$$\text{Total HP} = 242.5 \text{ HP}$$

To avoid the worst condition, there is a safety factor for the motor, as rule of thumb it is divided by 80%.

$$\text{Required motor HP} = \frac{\text{Total HP}}{80\%}$$

$$\text{Required motor HP} = \frac{242.5}{80\%}$$

$$\text{Required motor HP} = 303.2 \text{ HP}$$

From Required Motor HP Calculation we can choose the motor. So, from the catalog we choose 540 Series, 1 EA 63 HP-855 V-46A, 1 EA 250 HP-1237V-127.1A

Cable Voltage Drop

$$\text{Cable Voltage Drop} = (\text{PSD} + 100 \text{ ft}) \times (\text{Voltage Drop per 1000 ft})$$

$$\text{Cable Voltage Drop} = (6000 + 100 \text{ ft}) \times (36)$$

$$\text{Cable Voltage Drop} = 219.6 \text{ V}$$

Required Surface Voltage

$$\text{Required Surface Voltage} = \text{Motor Voltage} + \text{Cable Voltage}$$

$$\text{Required Surface Voltage} = 2,092 + 219.6$$

Required Surface Voltage = 2,331.6 V

KVA

$$KVA = \frac{\text{Surface Voltage} \times \text{Motor Ampere} \times 1.732}{1000}$$

$KVA = \frac{2,331.6 \times 173.1 \times 1.732}{1000}$
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KVA = 700 KVA